Problem 1)

a)

Assuming the voltages $x(t)$ and $y(t)$:
For RC:

$$i(t) = C \cdot \frac{\partial y(t)}{dt}$$

$$x(t) = R \cdot i(t) + y(t) = RC\dot{y}(t) + y(t)$$

For RL:

$$y(t) = L \cdot \frac{\partial i(t)}{dt}$$

$$x(t) = Ri(t) + L\dot{i}(t)$$

$$H_{RC}(s) = \frac{Y(s)}{X(s)} = \frac{\frac{1}{Cs}}{R + \frac{1}{Cs}} = \frac{1}{1 + RCs}$$

$$H_{RL}(s) = \frac{Y(s)}{X(s)} = \frac{Ls}{R + Ls} = \frac{s}{\frac{R}{L} + s} = \frac{1}{1 + \frac{R}{Ls}}$$

b)

$$H_{RC}(j\omega) = \frac{1}{1 + j\omega RC}$$

$$H_{RL}(j\omega) = \frac{j\omega}{\frac{R}{L} + j\omega} = \frac{1}{1 + \frac{R}{j\omega L}}$$

RC is a lowpass, while RL is a highpass.

c)

For RC $\tau = RC$ and for RL $\tau = \frac{L}{R}$
Unit pulse response is the given as:

$$h(t) = \frac{1}{\tau} e^{-t/\tau} u(t)$$

Problem 2)

a)

$$H(z) = \frac{1}{1 - \frac{2}{3} z^{-1}}$$

$$\implies h[n] = \left(\frac{2}{3}\right)^n u[n] \;\; \& \;\; \text{ROC: } |z| > \frac{2}{3}$$

b)

$$H(z) = \frac{1}{\left(1+\frac{1}{2}z^{-1}\right)(1-z^{-1})} = \frac{A}{\left(1+\frac{1}{2}z^{-1}\right)} + \frac{B}{(1-z^{-1})} \implies \frac{\frac{1}{3}}{1+\frac{1}{2}z^{-1}} + \frac{\frac{2}{3}}{1-z^{-}1}$$

$$\implies h[n] = \left(\frac{1}{3}\left(-\frac{1}{2}\right)^n + \frac{2}{3}\right)u[n] \quad \& \;\; \text{ROC: } |z| > 1$$

c)

$$H(z) = \frac{z^{-1}}{\left(1-\frac{3}{2}z^{-1}\right)(1-3z^{-1})} = \frac{A}{\left(1-\frac{3}{2}z^{-1}\right)} + \frac{B}{(1-3z^{-1})} \implies -\frac{2}{3}\left(\frac{1}{1-\frac{3}{2}z^{-1}} + \frac{1}{(1-3z^{-1})}\right)$$

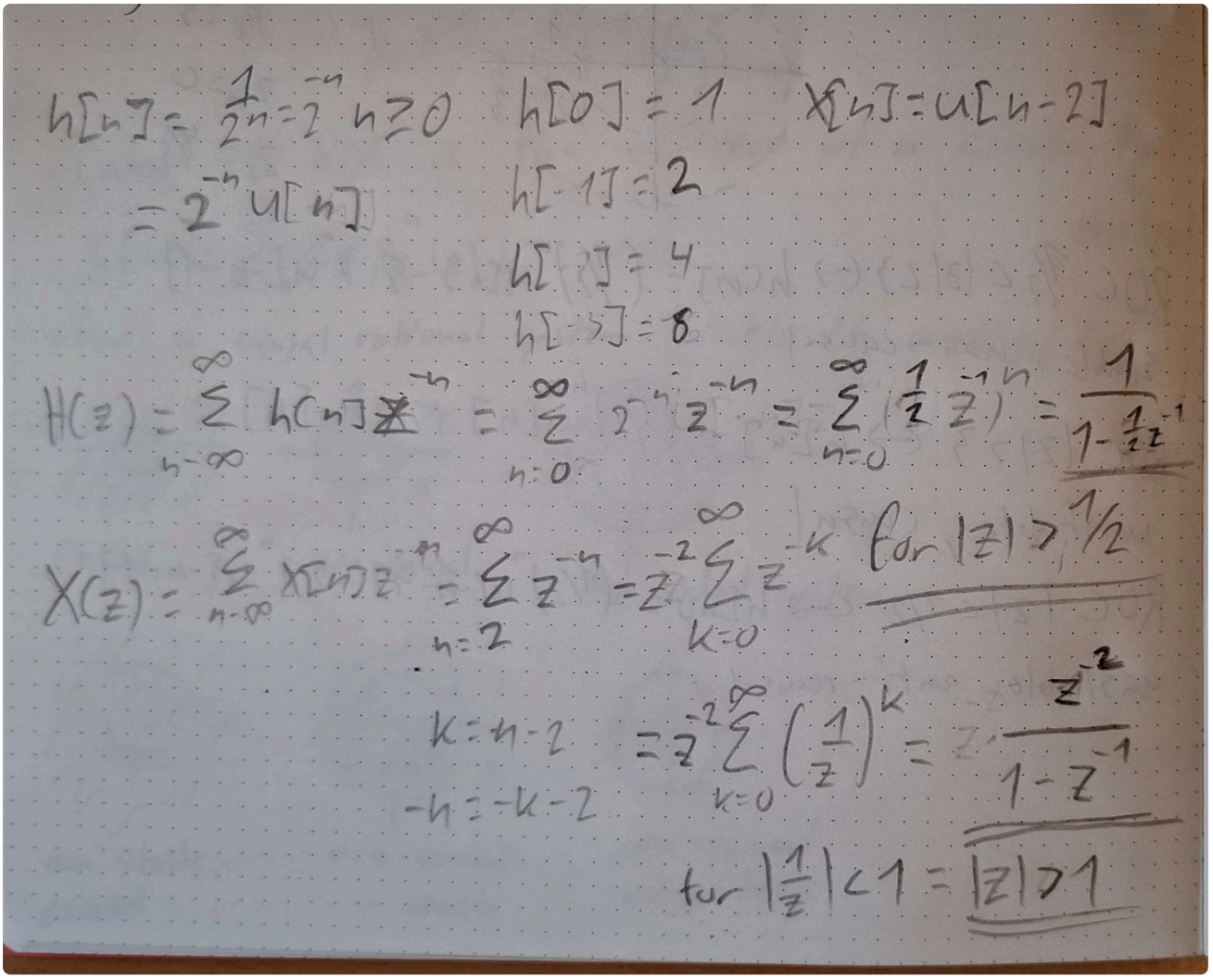
$$\implies h[n] = -\frac{2}{3}\left[-\left(\frac{3}{2}\right)^n - 3^n\right]u[-n-1] \;\; \& \;\; \text{ROC: } |z| < \frac{3}{2}$$

d)

For the system to be BIBO stable, the unit circle has to be within ROC, which goes for a) and c) as they are both true for $|z| = 1$.

Problem 3)

a)


h[n] = (1/2)^n = 2^-n n≥0
= 2^-n u[n]
h[0] = 1
h[1] = 2
h[2] = 4
h[3] = 8
x[n] = u[n-2]
H(z) = Σ_{n=-∞}^{∞} h[n] z^-n = Σ_{n=0}^{∞} 2^-n z^-n = Σ_{n=0}^{∞} (1/2 z^-1)^n = 1 / (1 - 1/2 z^-1)
for |z| > 1/2
X(z) = Σ_{n=-∞}^{∞} x[n] z^-n = Σ_{n=2}^{∞} z^-n = z^-2 Σ_{k=0}^{∞} z^-k
k = n-2
-n = -k-2
= z^-2 Σ_{k=0}^{∞} (1/z)^k = z^-2 / (1 - z^-1)
for |1/z| < 1 = |z| > 1

b & c)

b) $x[n] = u[n-2] \quad h[n] = \left(\frac{1}{2}\right)^n u[n]$

$$y[n] = \sum_{k=-\infty}^{\infty} u[k-2]\left(\frac{1}{2}\right)^{n-k} u[n-k]$$

for $k \geq 2$ and $k \leq n \Rightarrow y[n] = 0$ for $n < 2$

$$\Rightarrow \sum_{k=2}^{n}\left(\frac{1}{2}\right)^{n-k} = \left(\frac{1}{2}\right)^n \sum_{k=2}^{n}\left(\frac{1}{2}\right)^{-k} = \left(\frac{1}{2}\right)^n \sum_{k=2}^{n} 2^k$$

$$\Rightarrow 4\cdot\left(\frac{1}{2}\right)^n \sum_{k=0}^{n-2} 2^k = 4\left(\frac{1}{2}\right)^n \left(\frac{1-2^{(n-2)+1}}{1-2}\right) = 4\left(\frac{1}{2}\right)^n\left(2^{n-1}-1\right)$$

$$= 2 - 4\left(\frac{1}{2}\right)^n$$

$$\Rightarrow y[n] = \left[2 - 4\left(\frac{1}{2}\right)^n\right] u[n]$$

c) $X(z) = \frac{1}{\left(1-\frac{1}{2}z^{-1}\right)} \cdot \frac{z^2}{\left(1-z^{-1}\right)} \Rightarrow z^{-2} = A\left(1-z^{-1}\right) + B\left(1-\frac{1}{2}z^{-1}\right)$

$4 = A(1-2) \Rightarrow A = -4 \qquad 1 = B\left(1-\frac{1}{2}\right) \Rightarrow B = 2$

$$Y(z) = -\frac{4}{1-\frac{1}{2}z^{-1}} + \frac{2}{1-z^{-1}}$$

$$\Rightarrow y[n] = -4\left(\frac{1}{2}\right)^n u[n] + 2u[n] = \left[2 - 4\left(\frac{1}{2}\right)^n\right] u[n]$$

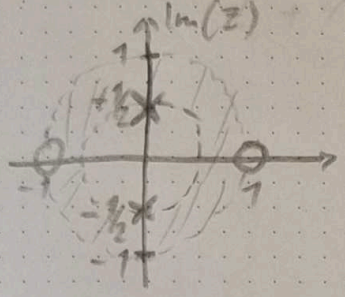
Problem 4)

a) $b_{k=0} = 1 \quad b_{k=1} = 0 \quad b_{k=2} = -1$

$a_{k=0} = 0 \quad a_{k=1} = 0 \quad a_{k=2} = 1/4$

$$\Rightarrow H(z) = \frac{1 - z^{-2}}{1 + 1/4\, z^{-2}} = \frac{(1+z)\cdot(1-z)}{(z+1/2j)(z-1/2j)}$$

b) Poles are in $z = \pm 1/2j$ and zeros in $z = \pm 1$



$z = e^{jw}$, go from $w[0, \pi]$

c) Since both poles are within the unit circle, and the filter is causal meaning the ROC is for $|z| > 1/2$. Thus the unit circle is within ROC and it's BIBO stable.

d) We can see that for low frequencies, we lie in $z = 1$ for $w = 0$ which is our zero. Going to $w = \pi/2$ we reach $z = j$ and are close to the top pole, amplifying these frequencies. For high freq $w = \pi$, $z = -1$ it's zero again. Therefore we have a bandpass filter